

## NPN SILICON TRANSISTORS 2SC3478, 2SC3478A

**DESCRIPTION** The 2SC3478/3478A is designed for general-purpose applications requiring high Breakdown Voltages.

### FEATURES

- High Breakdown Voltage.  
 $V_{CEO} = 180\text{ V}/200\text{ V}$  (2SC3478/2SC3478A)
- Good  $h_{FE}$  linearity.
- A complementary pair with 2SA1376/2SA1376A.

### ABSOLUTE MAXIMUM RATINGS

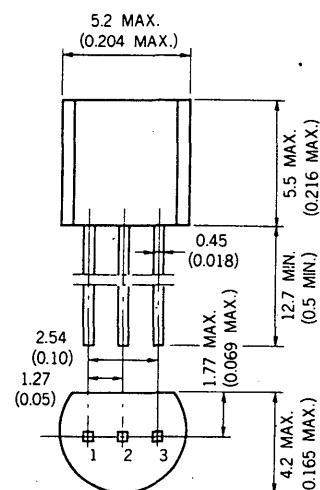
Maximum Temperatures  
 Storage Temperature . . . . .  $-55$  to  $+150\text{ }^{\circ}\text{C}$   
 Junction Temperature . . . . .  $150\text{ }^{\circ}\text{C}$  Maximum  
 Maximum Power Dissipation ( $T_a = 25\text{ }^{\circ}\text{C}$ )  
 Total Power Dissipation . . . . .  $750\text{ mW}$   
 Maximum Voltages and Currents ( $T_a = 25\text{ }^{\circ}\text{C}$ )

#### 2SC3478/2SC3478A

|           |                                    |         |    |
|-----------|------------------------------------|---------|----|
| $V_{CBO}$ | Collector to Base Voltage . . . .  | 200     | V  |
| $V_{CEO}$ | Collector to Emitter Voltage . .   | 180/200 | V  |
| $V_{EBO}$ | Emitter to Base Voltage . . . . .  | 5.0     | V  |
| $I_C$     | Collector Current (DC) . . . . .   | 100     | mA |
| $I_C$     | Collector Current (pulse)* . . . . | 200     | mA |
| $I_B$     | Base Current (DC) . . . . .        | 20      | mA |

\*PW  $\leq 10\text{ ms}$ , Duty Cycle  $\leq 50\%$

### PACKAGE DIMENSIONS in millimeters (inches)



1. EMITTER EIAJ : SC-43B  
 2. COLLECTOR JEDEC : TO-92  
 3. BASE IEC : PA33

### ELECTRICAL CHARACTERISTICS ( $T_a = 25\text{ }^{\circ}\text{C}$ )

#### 2SC3478/2SC3478A

| SYMBOL        | CHARACTERISTIC               | MIN. | TYP. | MAX.    | UNIT          | TEST CONDITIONS                                           |
|---------------|------------------------------|------|------|---------|---------------|-----------------------------------------------------------|
| $h_{FE}$      | DC Current Gain              | 135  |      | 400/600 | —             | $V_{CE} = 10\text{ V}$ , $I_C = 10\text{ mA}$             |
| $t_{on}$      | Turn-on Time                 |      | 0.15 |         | $\mu\text{s}$ | $I_C = 10\text{ mA}$                                      |
| $t_{off}$     | Turn-off Time                |      | 1.6  |         | $\mu\text{s}$ | $I_{B1} = -I_{B2} = 1\text{ mA}$ , $V_{CC} = 10\text{ V}$ |
| $f_T$         | Gain Bandwidth Product       | 100  | 150  |         | MHz           | $V_{CE} = 10\text{ V}$ , $I_E = -10\text{ mA}$            |
| $C_{ob}$      | Output Capacitance           |      | 2.6  | 3.5     | pF            | $V_{CB} = 30\text{ V}$ , $I_E = 0$ , $f = 1.0\text{ MHz}$ |
| $I_{CBO}$     | Collector Cutoff Current     |      |      | 100     | nA            | $V_{CB} = 200\text{ V}$ , $I_E = 0$                       |
| $I_{EBO}$     | Emitter Cutoff Current       |      |      | 100     | nA            | $V_{EB} = 4.0\text{ V}$ , $I_C = 0$                       |
| $V_{BE}$      | Base to Emitter Voltage      | 600  | 660  | 700     | mV            | $V_{CE} = 10\text{ V}$ , $I_C = 10\text{ mA}$             |
| $V_{CE(sat)}$ | Collector Saturation Voltage |      | 0.1  | 0.3     | V             | $I_C = 50\text{ mA}$ , $I_B = 5.0\text{ mA}$              |
| $V_{BE(sat)}$ | Base Saturation Voltage      |      | 0.8  | 1.2     | V             | $I_C = 50\text{ mA}$ , $I_B = 5.0\text{ mA}$              |

### Classification of $h_{FE}$

| Rank  | L         | K         | U**       |
|-------|-----------|-----------|-----------|
| Range | 135 — 270 | 200 — 400 | 300 — 600 |

Test Conditions:  $V_{CE} = 10\text{ V}$ ,  $I_C = 10\text{ mA}$

\*\* 2SC3478A has no U rank.

TYPICAL CHARACTERISTICS ( $T_a = 25^\circ\text{C}$ )

